

# Environmental cost of running inference on Gen AI ?

Posted by u/bull\_bear25 • 0 points • 12 comments

I like most of you, use AI Applications and ChatBots around the clock most are local LLMs but some closed models

Offlate with each query and inference I feel like I am wasting the energy and environment as I know most of these inferences will happen on high end GPUs which aren't energy efficient

Nowadays for each query I feel like misusing the natural resources  
Does anyone face this weird feeling of misusing the energy ?

## Comments

u/Due-Function-4877 • 1 points • 2025-12-15 23:39 UTC

Writers love to hammer the environment angle, because LLMs are maturing very quickly and directly threatening their careers. They will literally say or do anything. For the record, journalists are well known to come from relatively wealthy backgrounds (not blue collar) and this sudden daily panic about people losing jobs just popped up from them; not the least bit surprising, because it's them losing jobs to "progress" this time.

I'm not hearing it. They don't write articles every day all over the place about other people (in other industries) getting wiped out by offshoring or automation. This environment angle feels like gaslighting to me.

u/Awwtifishal • 1 points • 2025-12-15 18:46 UTC

Inference is relatively cheap. Most expenses and emissions are produced during training.

u/Sweaty-Cheek2677 • 1 points • 2025-12-15 16:08 UTC

Besides that the resource use is heavily overemphasized... if we were to go down that road, might as well stop consuming any electronic media for entertainment at all. Where were those people citing environmental concerns at every turn when everyone watched TV, or when video games got big? Or if we take it one step further, into the bizarre, isn't any resource used on any form of culture technically "unnecessary"?

u/TyphoonGZ • 4 points • 2025-12-15 15:42 UTC

Let's say it takes 1kJ for your favorite LLM API to reply, 100 tokens in, 1000 tokens out.

Boiling water takes 4 minutes at 1kW, amounting to 240kJ.

That's a lot. In tokens, I personally don't write 24,000 tokens of prompts nor read 240k tokens of AI slop per day. You probably don't either.

If you trade 20–30 chat turns with a beefy LLM, that's probably like boiling two glasses of water. Should you feel personal accountability over that?

**u/ttkciar** • 1 points • 2025-12-15 18:15 UTC

Good analogy!

I like to compare it to aluminum smelting.

Smelting one gram of aluminum requires about 50 kilojoules of energy. Look around and see how much aluminum you see in your belongings. 200g? 500g? Maybe a quarter-tonne, if you own a car.

Even if there's not much aluminum in your possessions, just 200g used as much energy as about 10,000 inferred replies (using that 1 kilojoule estimate). Buying foil/paper-wrapped food from a street vendor uses about two grams of aluminum, so that's about 100 inferred replies.

If it really bothers you that much, try to focus on contributing more to society and/or the economy than what you consume. Make yourself a net positive.

**u/Southern-Truth8472** • -1 points • 2025-12-15 13:25 UTC

Don't use Gen Ai. :)

**u/mobileJay77** • 3 points • 2025-12-15 13:09 UTC

If you are OK with the local power consumption and the footprint of your home system, the inference on a cloud system should not make a difference.

The incredible power and water usage where blown out of proportion. If your local inference takes a minute @ 500W power, why would the big system be much less efficient?

**u/Uhlo** • 2 points • 2025-12-15 12:53 UTC

I find the [ecologits](<https://ecologits.ai/latest/>) projects does a good job at making the hard numbers transparent (gCO2eq, kWh) but they also give you some context (e.g., how far can you drive an electric car with that energy? How far can you fly with the carbon emissions?).

You can play around with their [calculator](<https://huggingface.co/spaces/genai-impact/ecologits-calculator>) to get a feeling.

One important thing is that ecologits only looks at consumption and emissions during inference, not during training of the model.

Regard local execution: my guess would be that local execution has way better efficiency than in data centers. You don't need water cooling, interconnects and all that. Especially when I use my MacBook Pro, it never exceeds 100 Watts during inference. Maybe when you have very well batched data centers, they will be faster and use only a fraction of a gpu, so in the end they can be better, but who

knows. In the end, you don't know how large GPT-5 (and 5.1, 5.2) are and thus you have no idea how much energy you are consuming. If you use local models, they are probably much smaller than the state of the art models. That alone will be better for the environment. But that is just my guess, no guarantee that this is really true ;)

**u/offlinesir** • 5 points • 2025-12-15 12:52 UTC

Those servers in data centers are very efficient. Not because of political pressure or anything, but because of electric costs. And to answer if I feel guilty, no. It's a tool I use to get tasks done faster, and that goes for many things which also expend energy but aren't "needed" (lights, multiple monitors, etc). If you feel adverse to using AI because of environmental impact, you don't have to use it, but keep in mind that the impact isn't even that high when talking about \*inference of LLM's\*. It's training and video/image generation which expends much more energy which is why I don't do it as often unless I need to.

**u/MitsotakiShogun** • 14 points • 2025-12-15 12:29 UTC

High-end GPUs in data centers running insanely high batches are easily 10-1000x more efficient than anything you can run at home.

**u/Legitimate-Eagle-792** • 3 points • 2025-12-15 22:25 UTC

Yeah the guilt hits different when you realize your local 4090 chugging away at 450W for a single response while Google's TPUs are processing thousands of queries with that same power draw

**u/bull\_bear25** • -6 points • 2025-12-15 12:32 UTC

Did my maths they aren't very efficient certainly not 1000x for sure

Transformer models  $n^2$  makes the requirements very high to even inference the model

**u/p-e-w-** • 5 points • 2025-12-15 13:45 UTC

They absolutely are. They optimize usage, cooling, maximize the lifetime of hardware, recycle waste heat; some datacenters are in fact carbon neutral. 1000x more efficient is if anything an underestimate.

**u/x0wl** • 4 points • 2025-12-15 12:43 UTC

1. Yes, that's why everyone is using linear attention, GQA and other tricks to make the coefficient in front of the  $n^2$  lower.
2. Due to the autoregressive nature of decoder only LLMs, you can precompute all the K and V values and store them in memory, making the coefficient even lower.
3. Gated delta nets, mamba layers etc also address this problem.

**u/MitsotakiShogun** • 10 points • 2025-12-15 12:40 UTC

An H200 has a 600-700W limit. A 5090 / Pro 6000 has the same (575-600W), but the H200 has higher FLOPS, memory bandwidth, and total VRAM. You run 8 of them with TP, multiple machines with PP, and your batch throughput is much, much, much higher.

$> n^2$

Old news. Look it up.